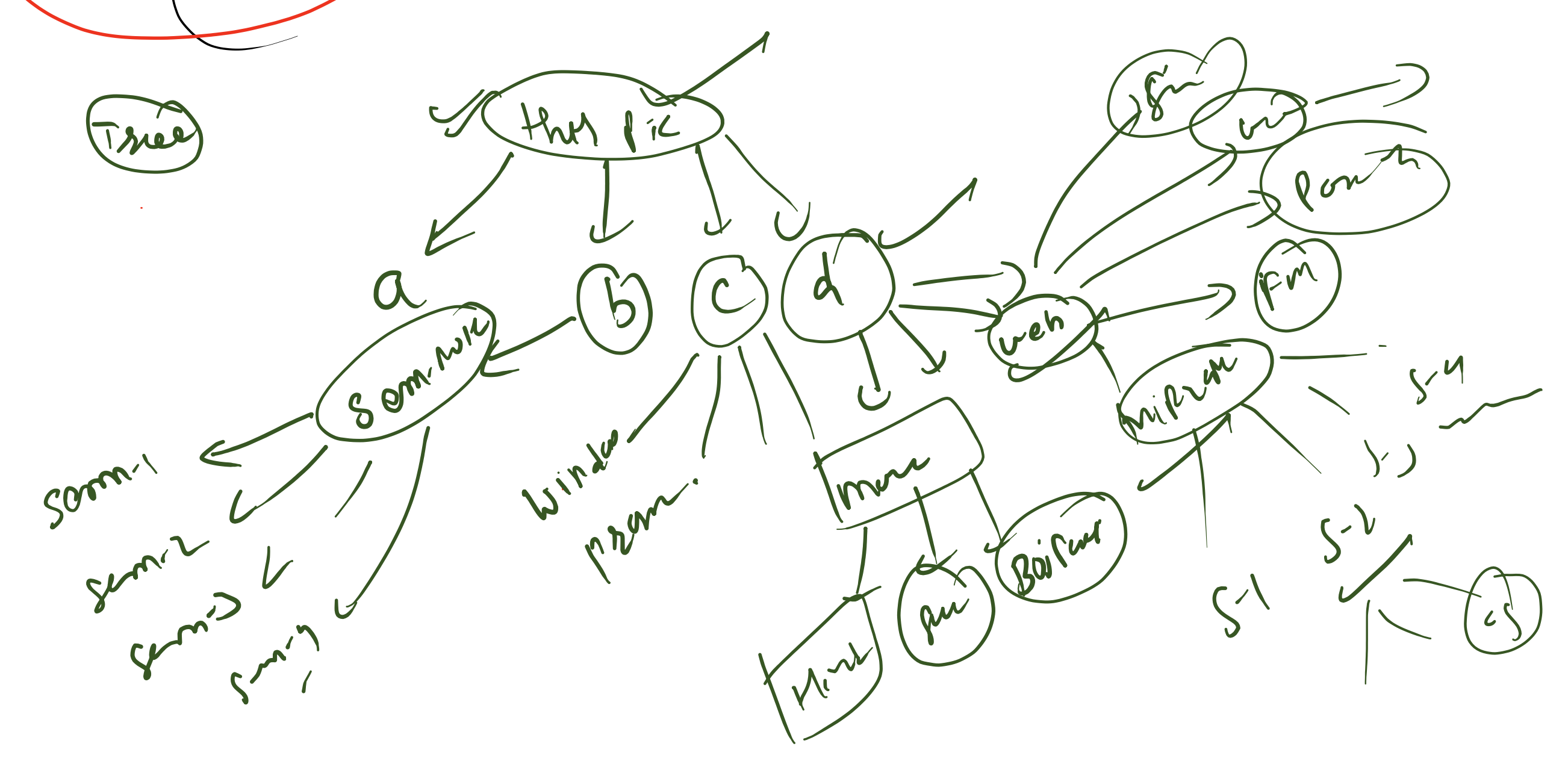
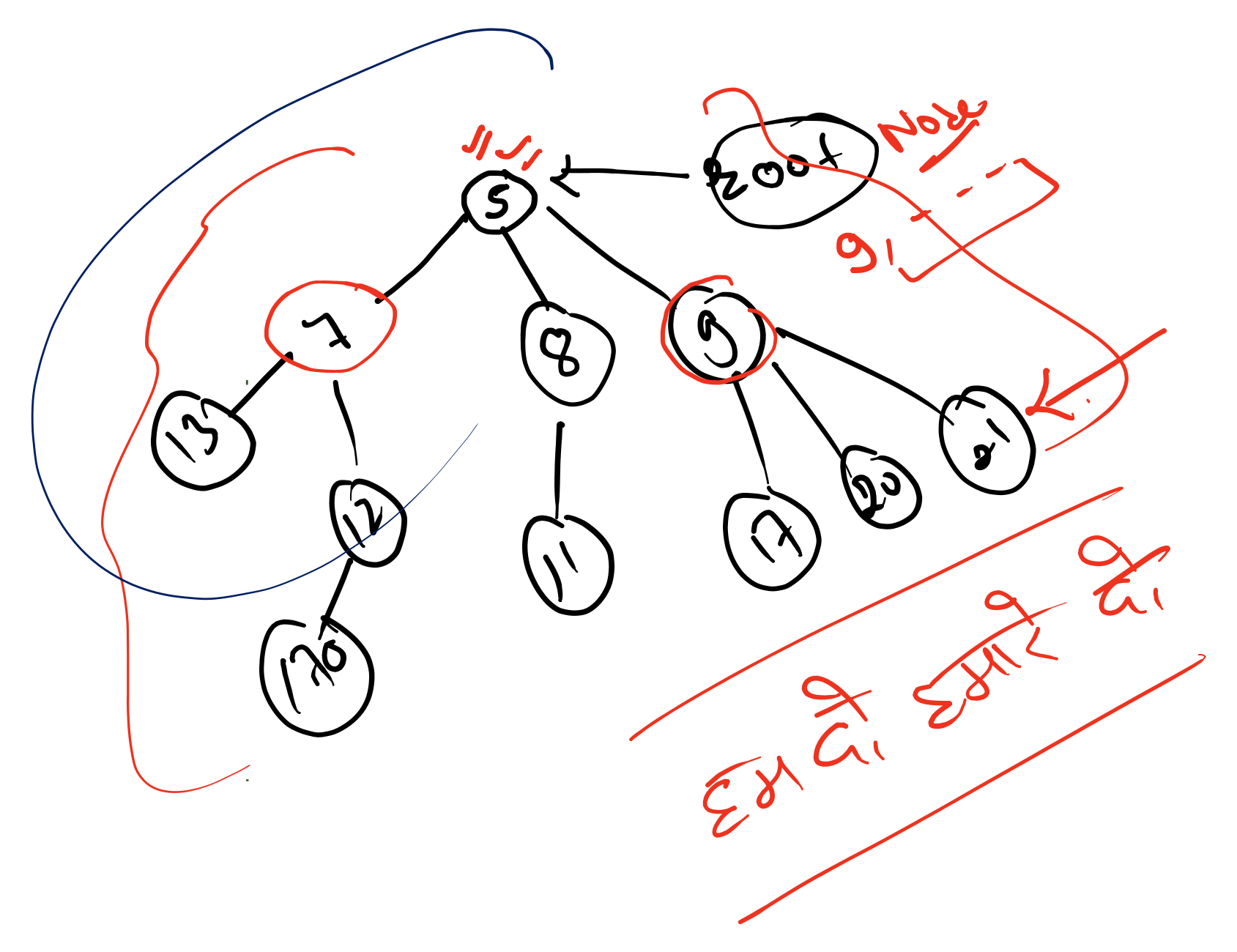
Lec-29
22/02/26 12:54 PM

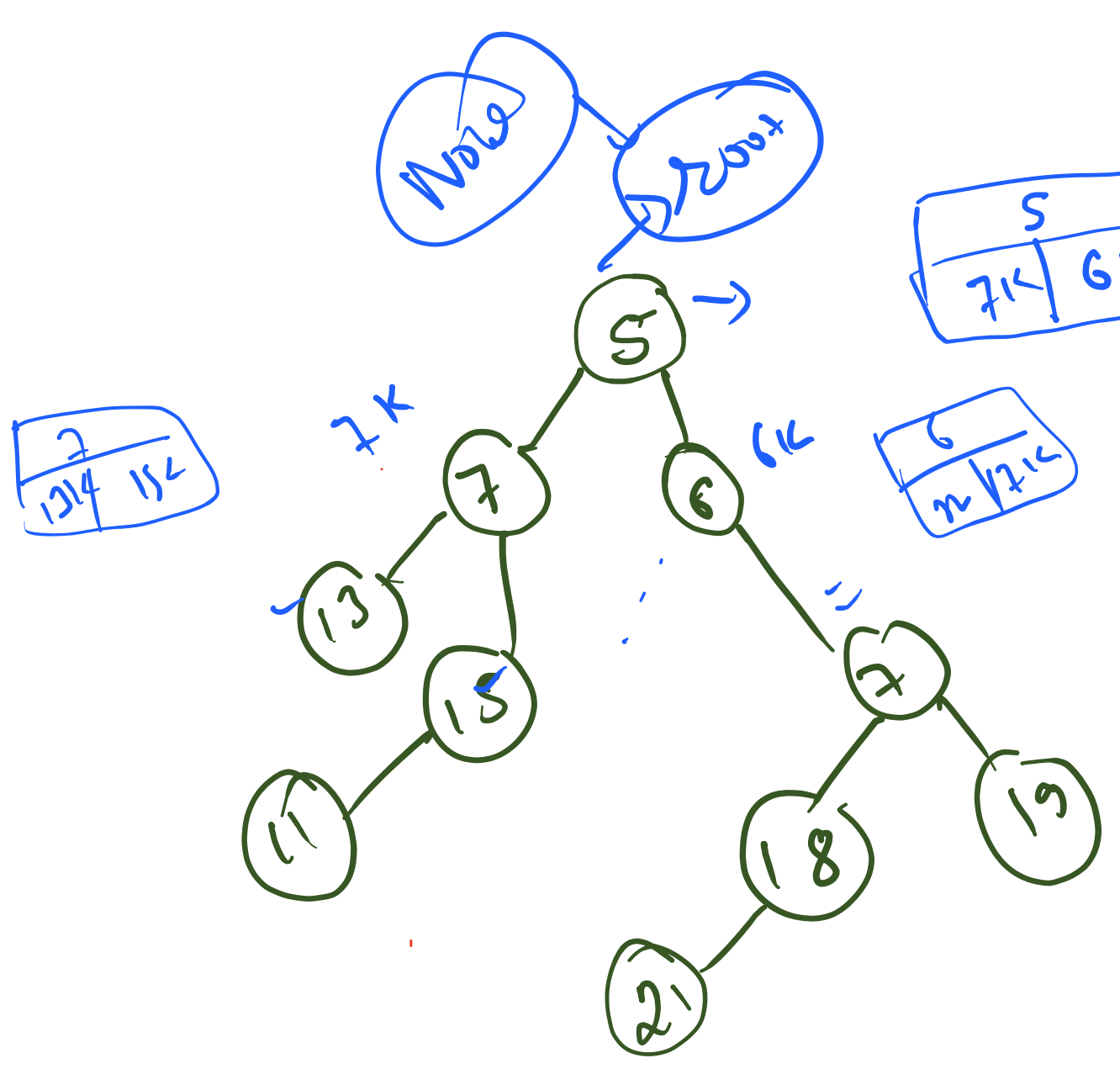
① Generic tree (GT)
Binary Tree BT
Binary Search Tree BST
Prefix Tree PT


Tree
My pc
a
b
c
d
Sem.work
Sem-1
Sem-2
Sem-3
Sem-4
window
program
movie
Hindi
web
Python
S-4
S-3
S-2
S-1

M-S-2-E8

add new

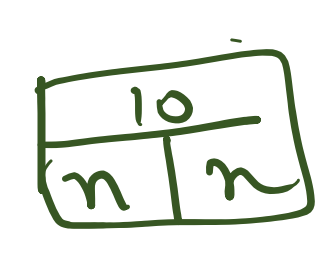

root
Node
9
5
7
8
9
13
12
11
17
20
21
170

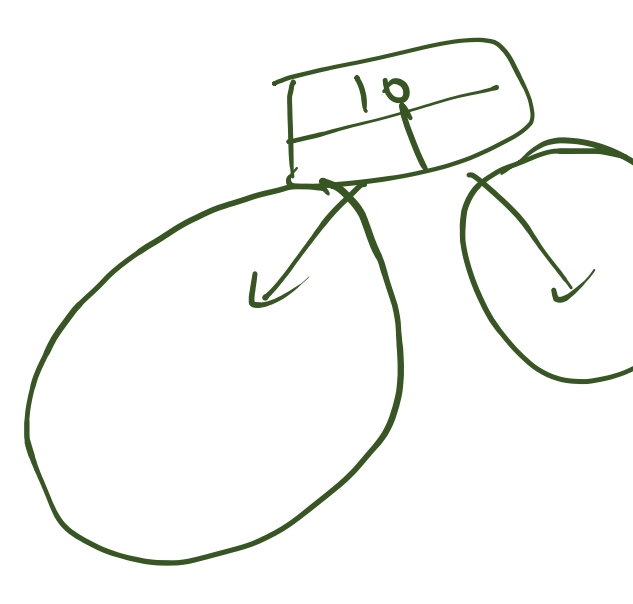

Node
root
5
7k 6k
6
n 7k
7
13 15
7k
6k
13
15
11
18
19
21

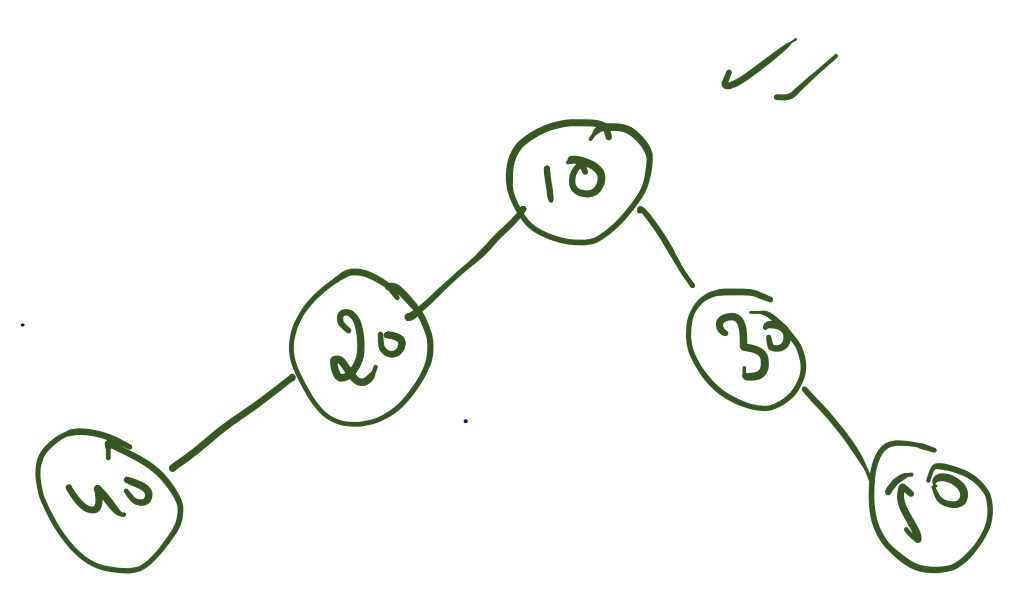
Class Node{
int val;
Node Left
Node Right

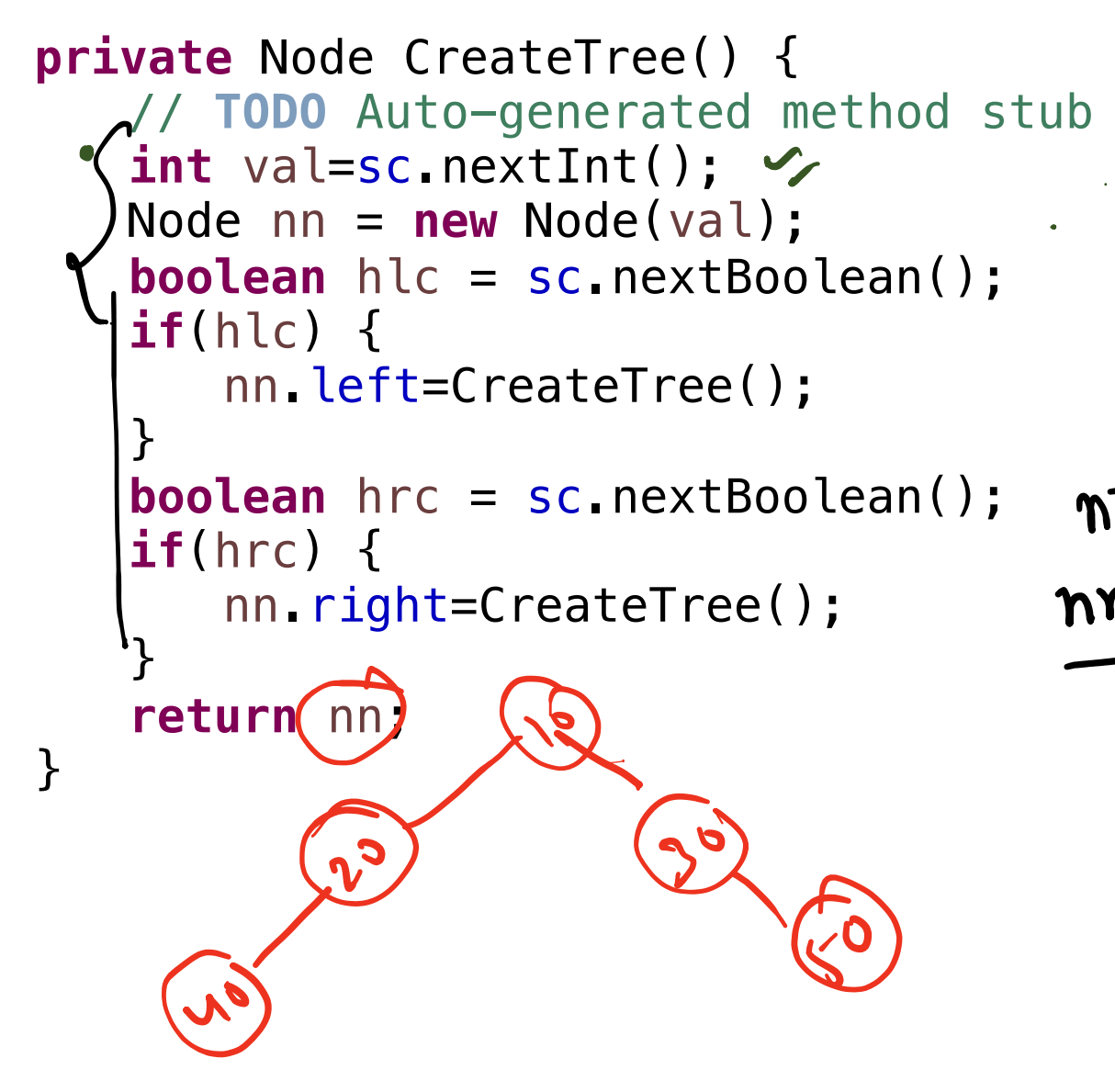
Root Left Right

[10 T 20 T 40 F F F T 30 F T 50 F F] → input






10
20
30
40
50

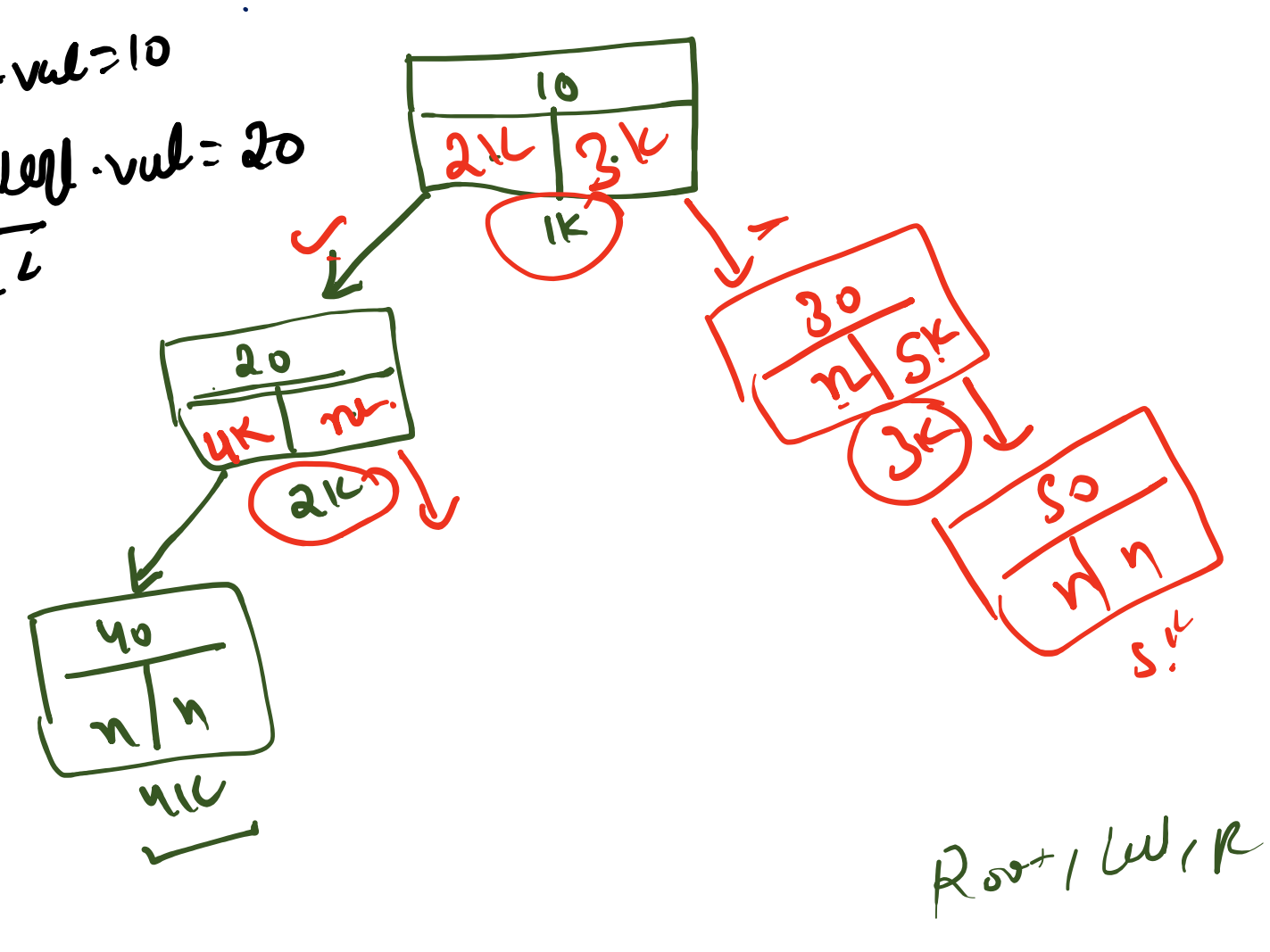


```
private Node CreateTree() {
    // TODO Auto-generated method stub
    int val=sc.nextInt();
    Node nn = new Node(val);
    boolean hlc = sc.nextBoolean();
    if(hlc) {
        nn.left=CreateTree();
    }
    boolean hrc = sc.nextBoolean();
    if(hrc) {
        nn.right=CreateTree();
    }
    return nn;
}
```

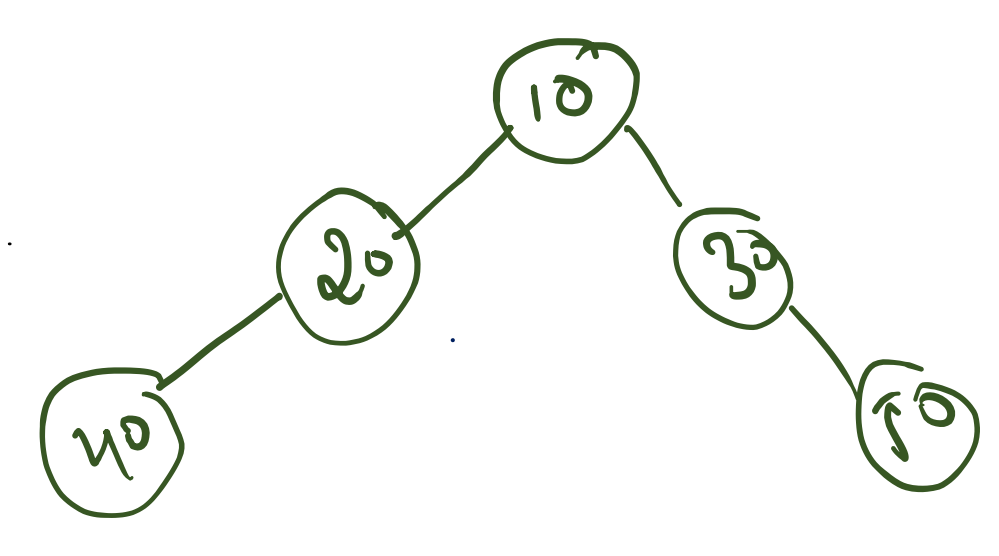
10
20
30
40
50

[10 T 20 T 40 F F F T 30 F T 50 F F] → input

nn.val=10
nn.left.val=20
2k

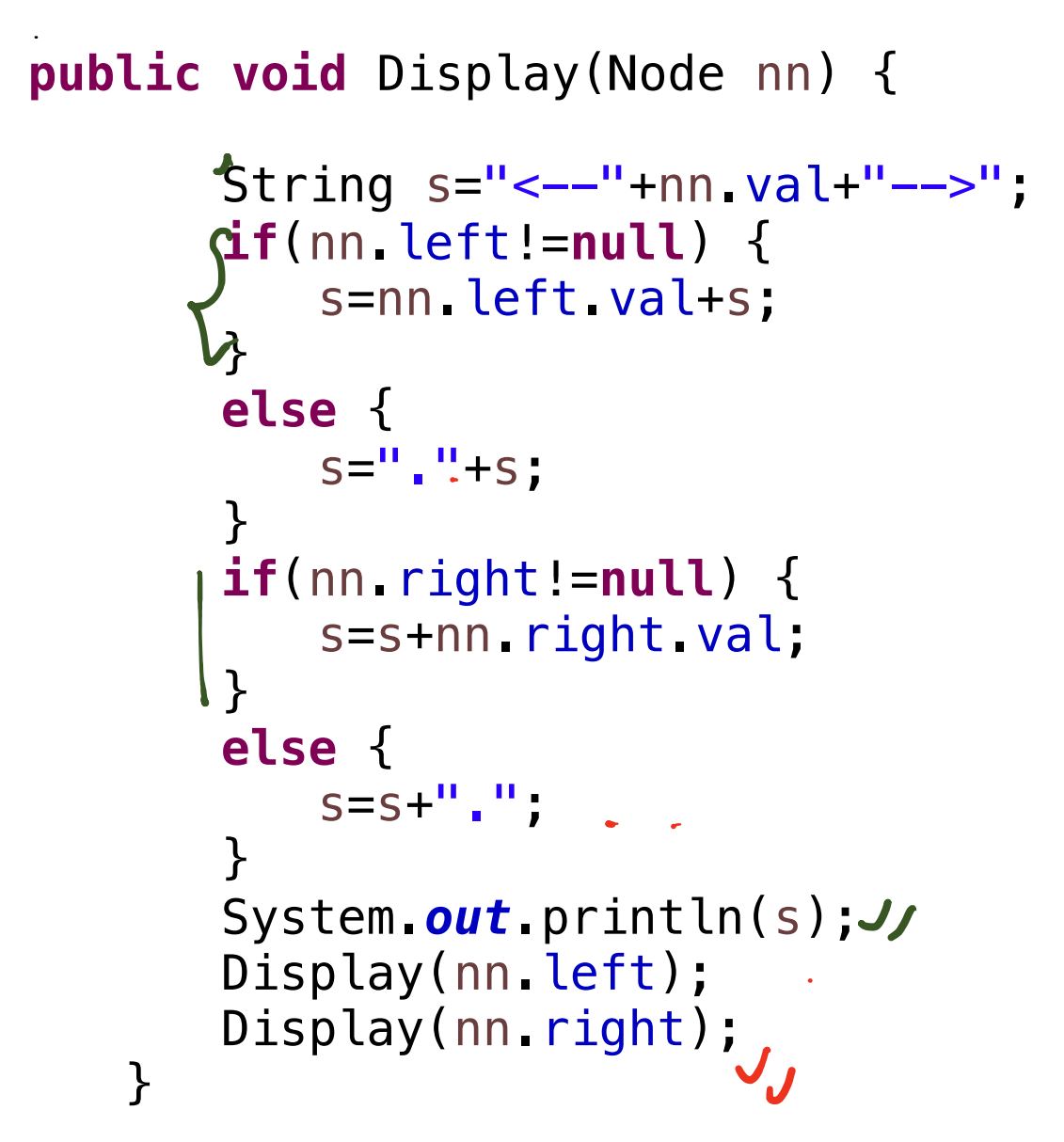

10
2k 3k
1k
20
4k n
2k
40
n n
4k
30
n 5k
3k
50
n n
5k

Root, Left, Right


10
20
30
40
50

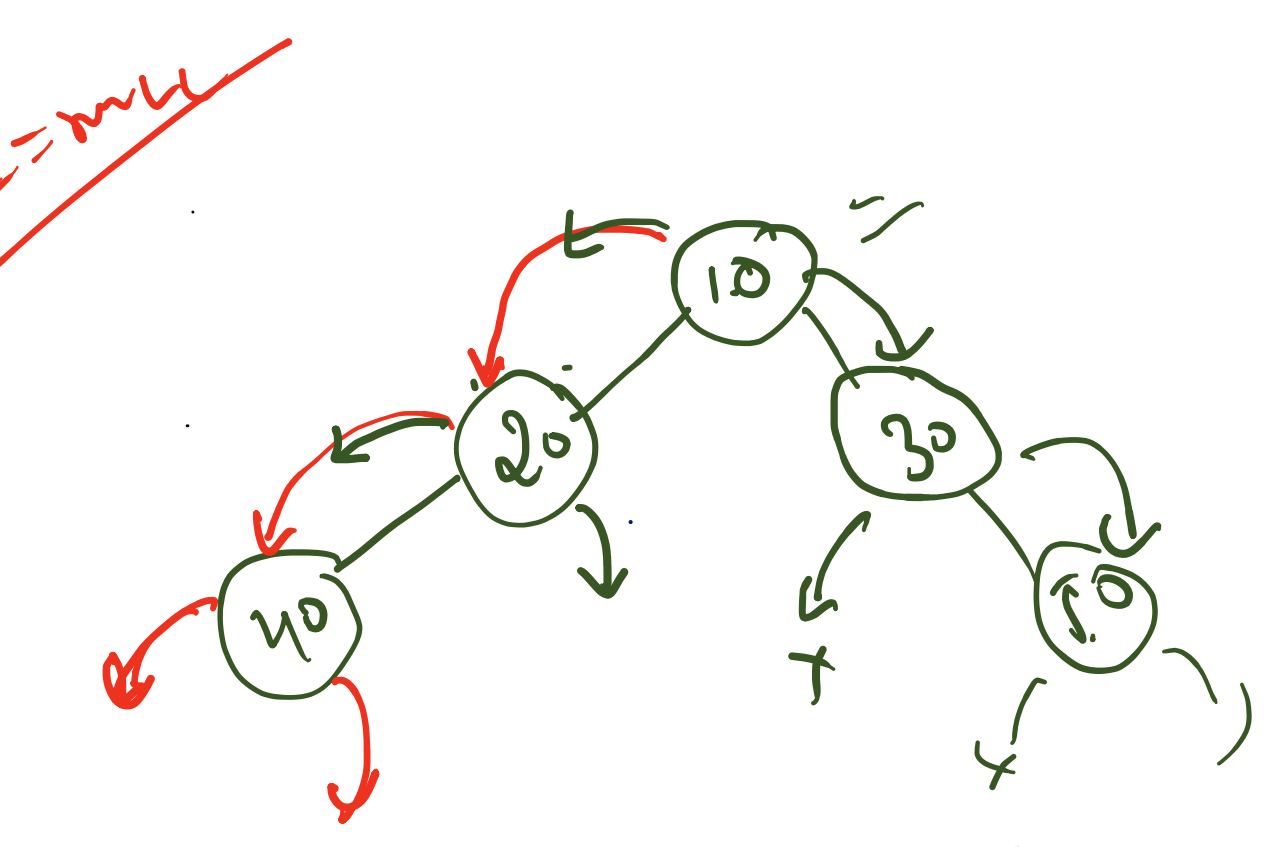
20 ← 10 → 30
40 ← 20 → .
. ← 40 → .
. ← 30 → 50
. ← 50 → .

Left ←10→
[ ]+s →

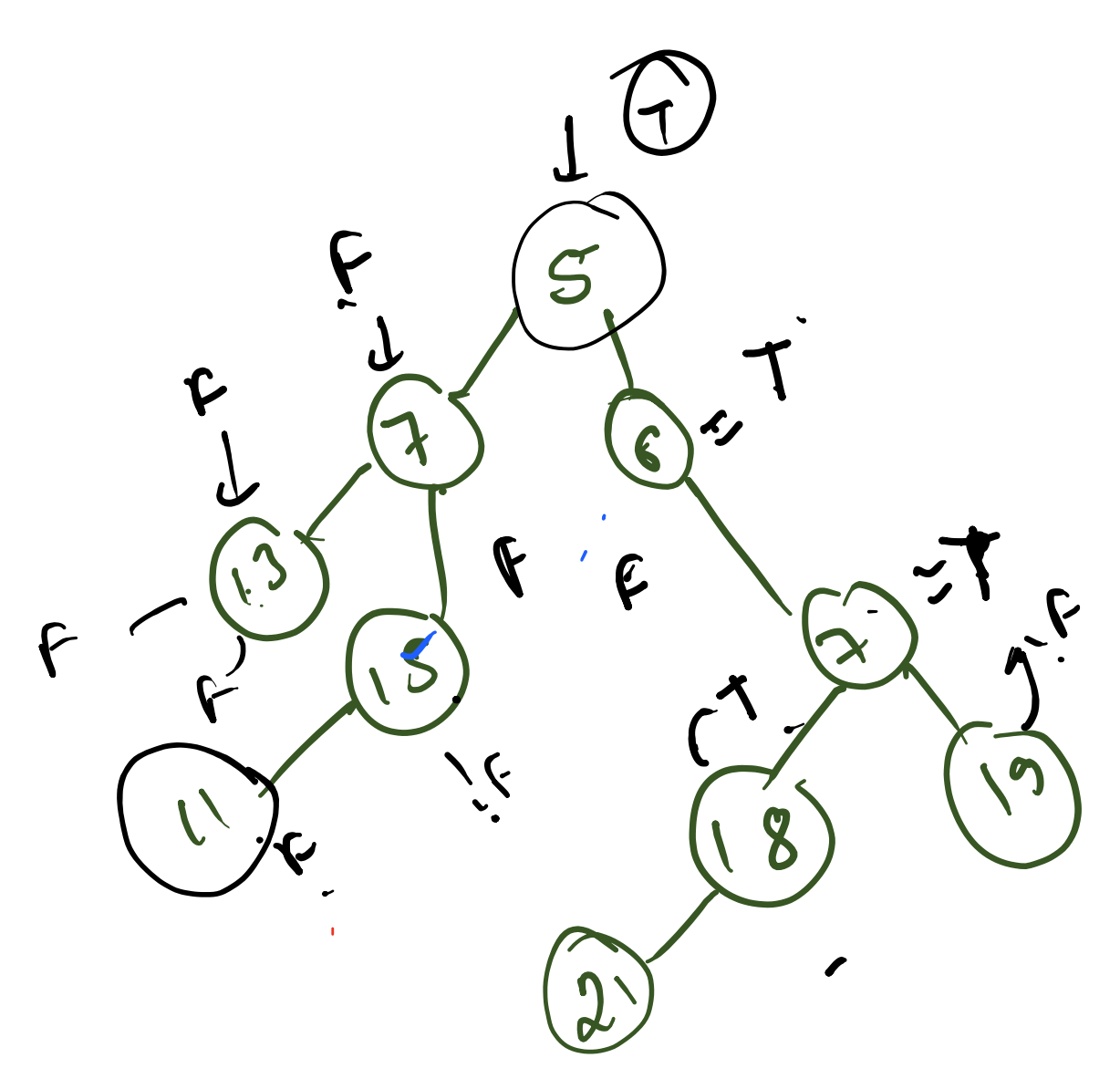


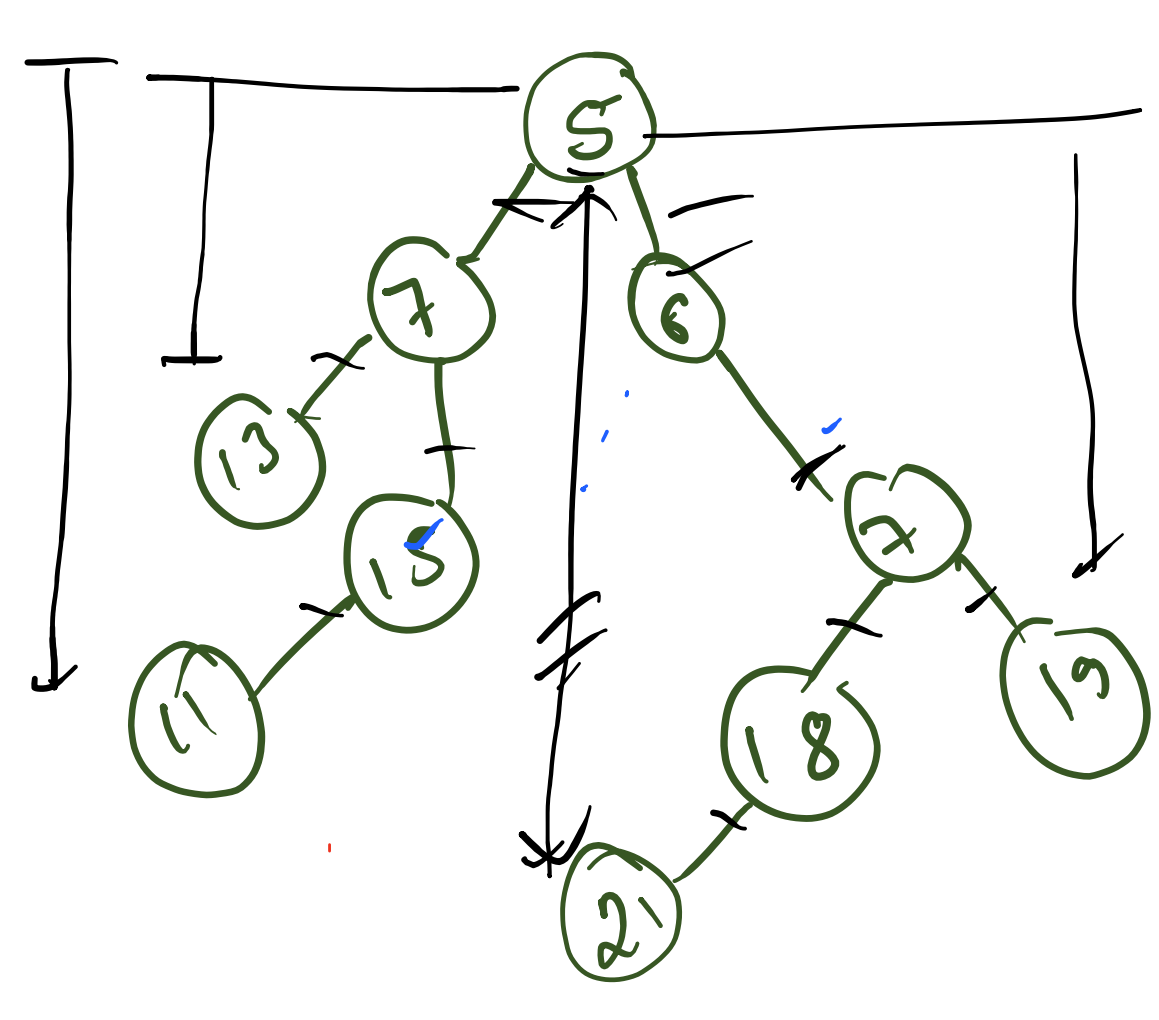
```
public void Display(Node nn) {

    String s="<--"+nn.val+"-->";
    if(nn.left!=null) {
        s=nn.left.val+s;
    }
    else {
        s="."+s;
    }
    if(nn.right!=null) {
        s=s+nn.right.val;
    }
    else {
        s=s+".";
    }
    System.out.println(s);
    Display(nn.left);
    Display(nn.right);
}
```

nn==null

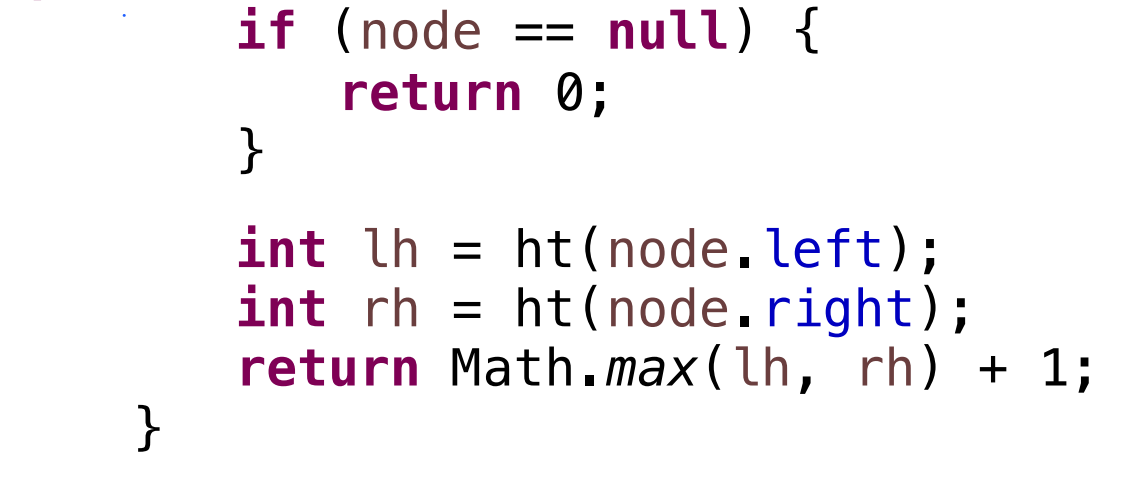

10
20
30
40
50

20 ← 10 → 30
40 ← 20 → .
. ← 40 → .
. ← 30 → 50
. ← 50 → .


T
5
F
7
6
T
F
13
15
F
F
7
T
F
11
18
19
21

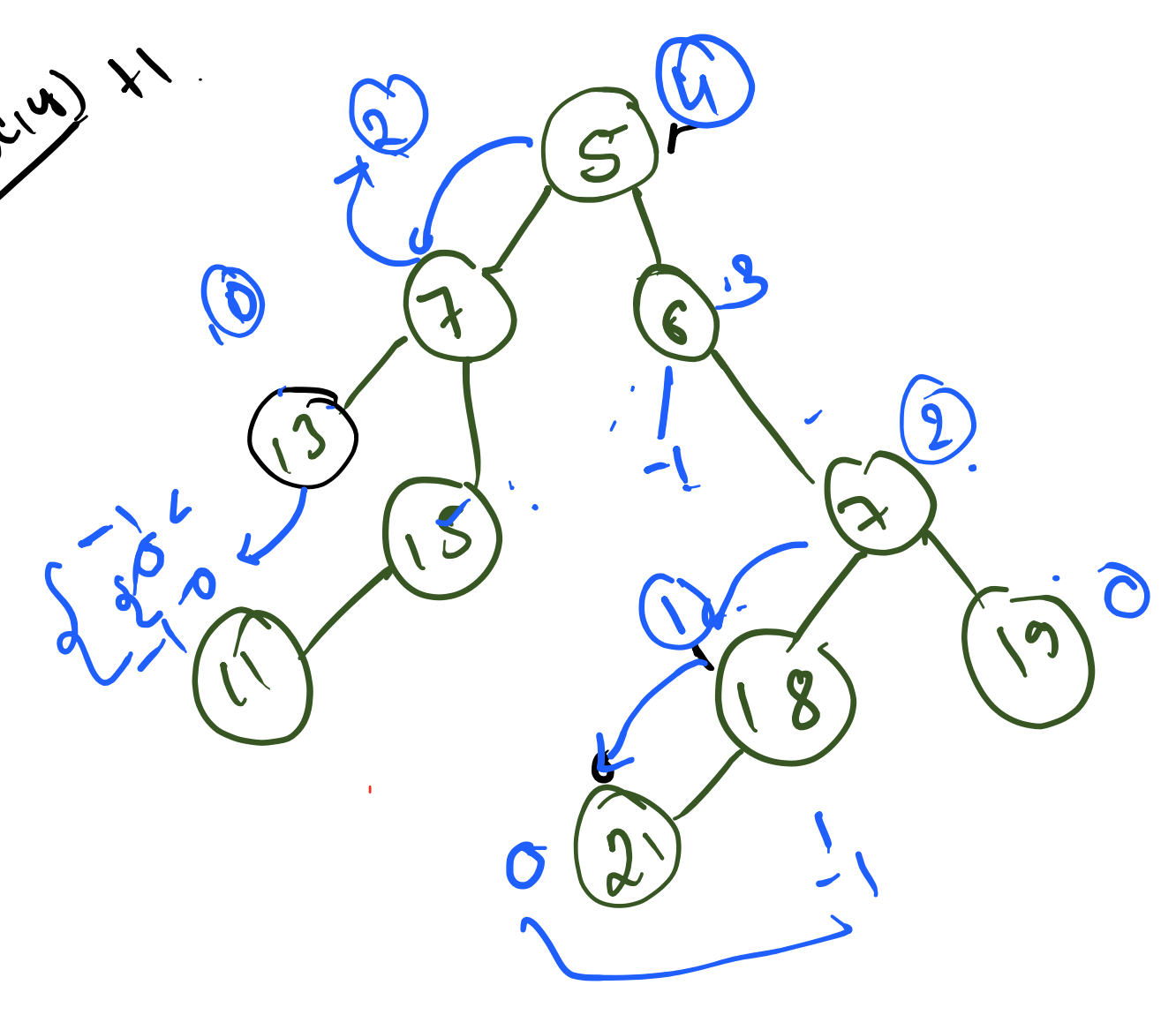

5
7
6
13
15
7
11
18
19
21

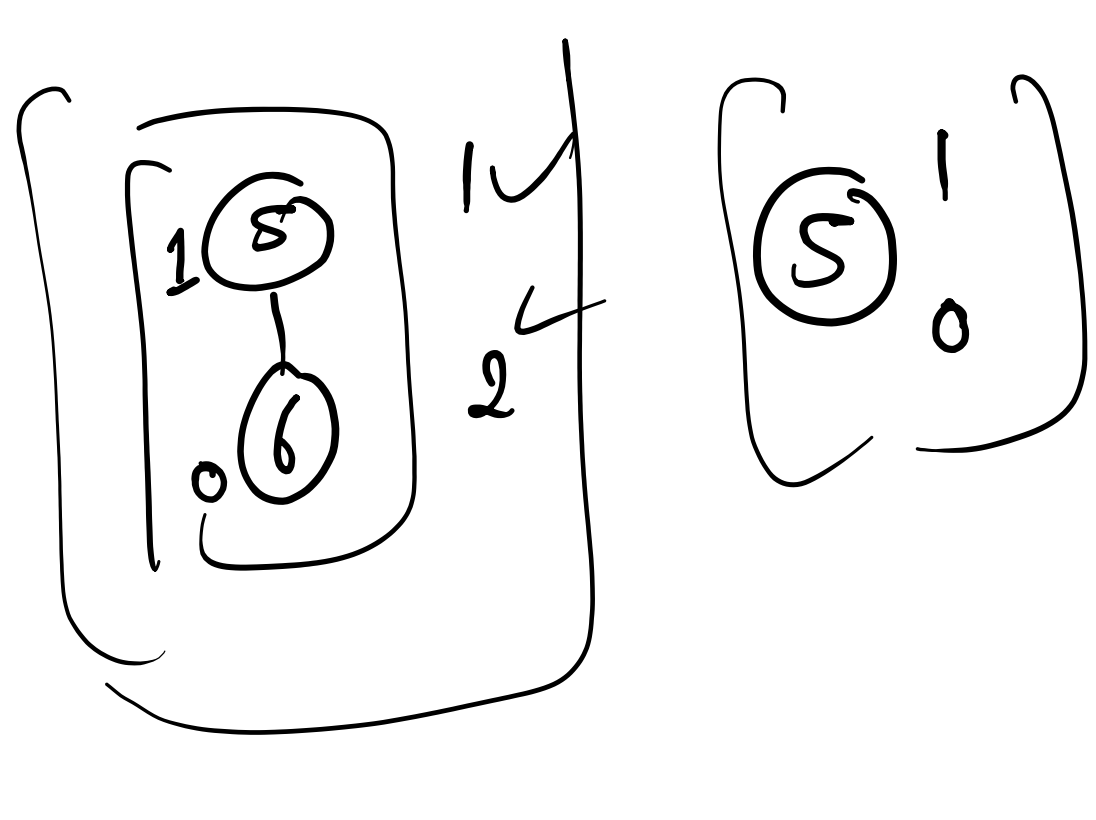
maximum distance b/w root to leaf



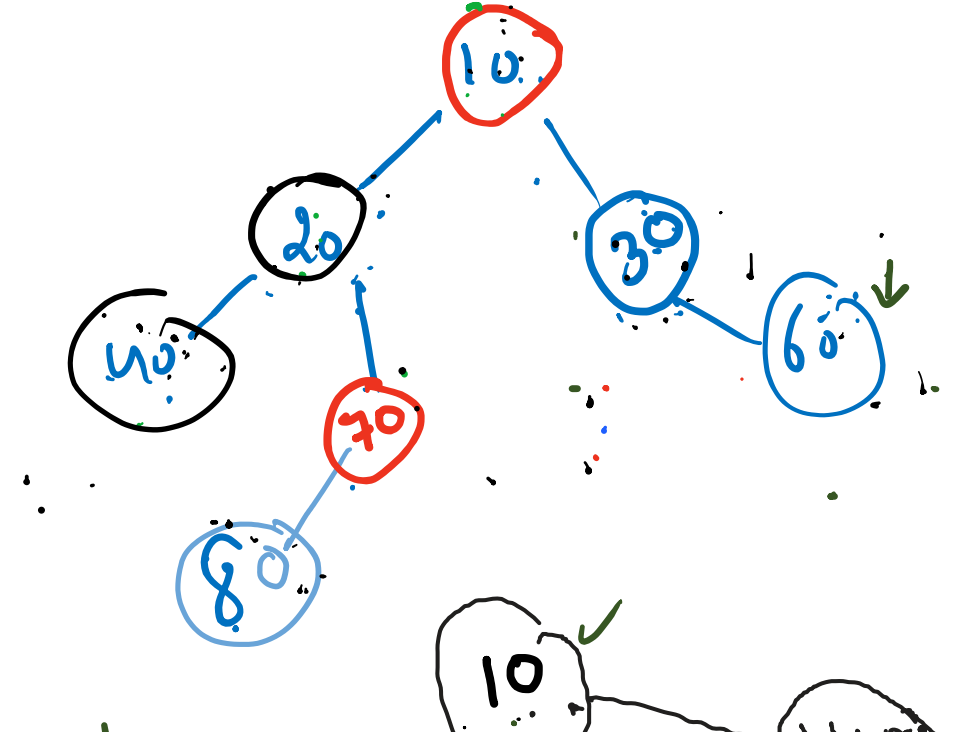
```
private int ht(Node node) {
    if (node == null) {
        return 0;
    }

    int lh = ht(node.left);
    int rh = ht(node.right);
    return Math.max(lh, rh) + 1;
}
```

max(x,y) +1


2
4
5
0
7
6
3
13
15
1
2
11
7
1
18
19
0
0
21
1



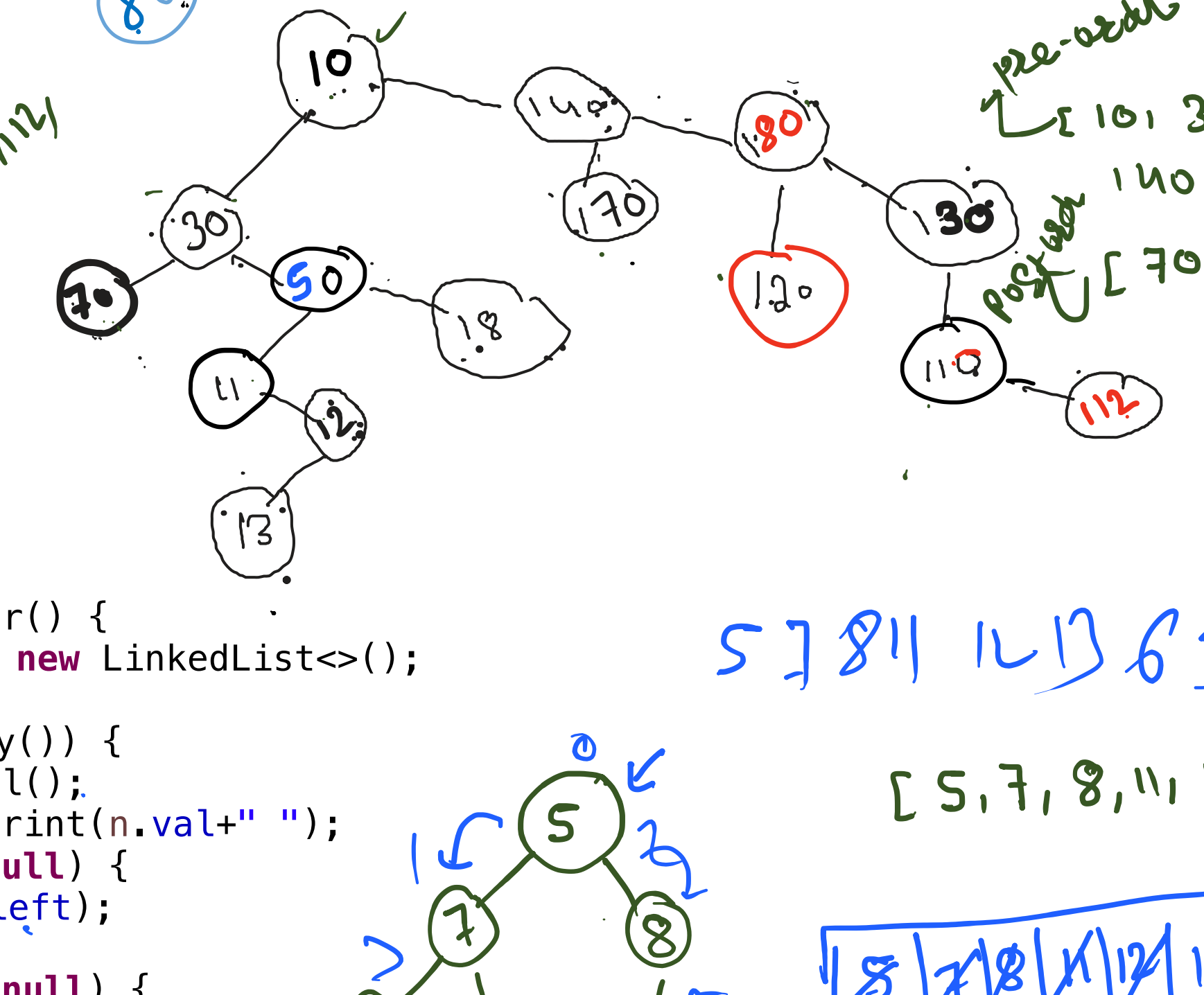
in-order
[40, 20, 80, 70, 10, 30, 60]


10
20
30
40
70
60
80

Pre-Order → Root Left Right
Post-Order → Left Right Root
In-order → Left Root Right

[10, 20, 40, 70, 80, 30, 60] pre-order

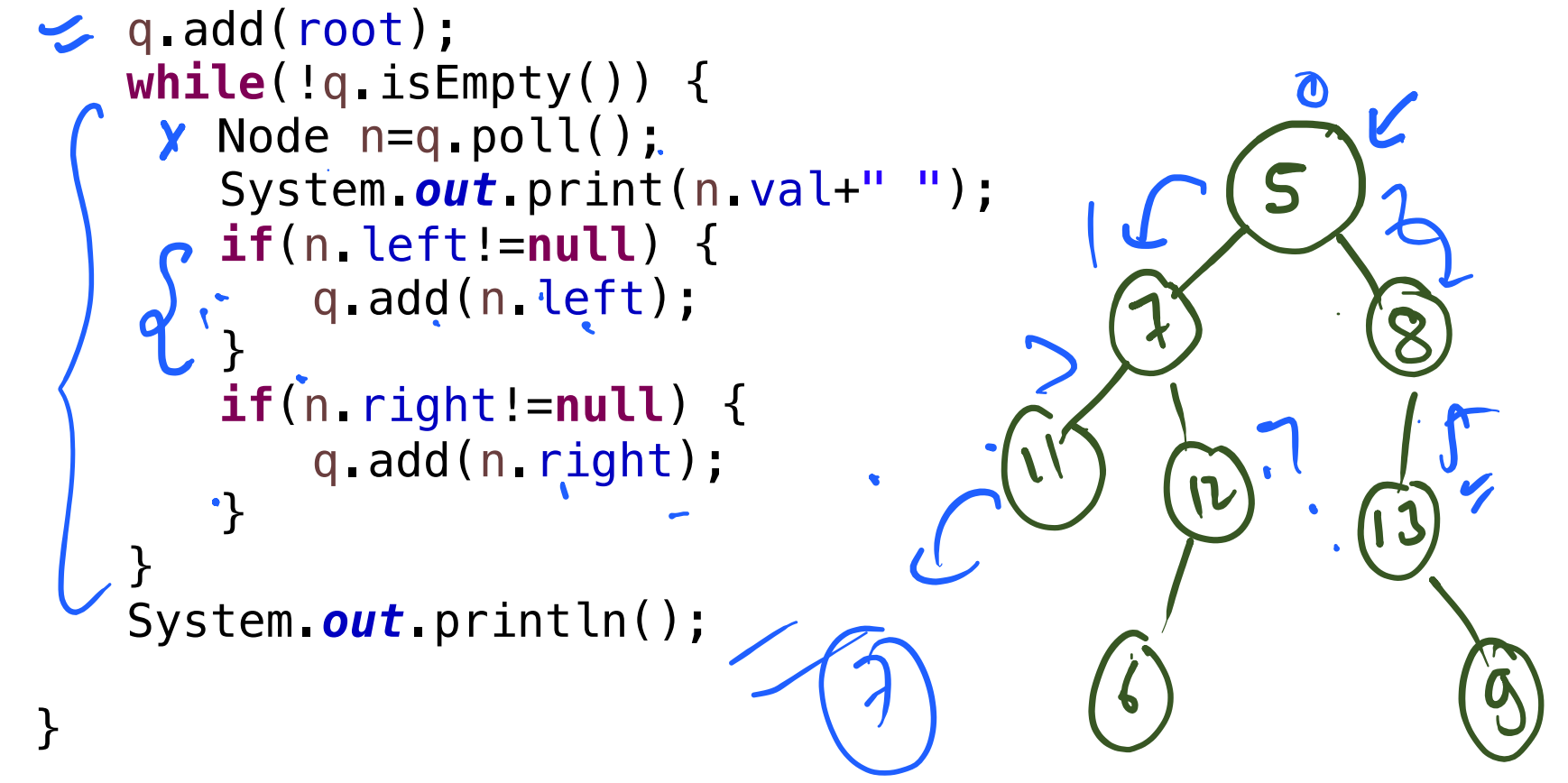
[40, 80, 70, 20, 60, 30, 10] post-order


10
30
70
50
11
12
13
18
140
170
180
120
130
110
112

pre-order
[10, 30, 70, 50, 11, 12, 13, 18,
140, 170, 180, 120, 130, 110, 112]

post-order
[70, 13, 12, 11, 18, 50, 30, 170, 120,
112, 110, 130, 180, 140, 10]

[70, 30, 11, 13, 12, 50, 18,
10, 170, 140, 120, 180, 110, 112,
130]

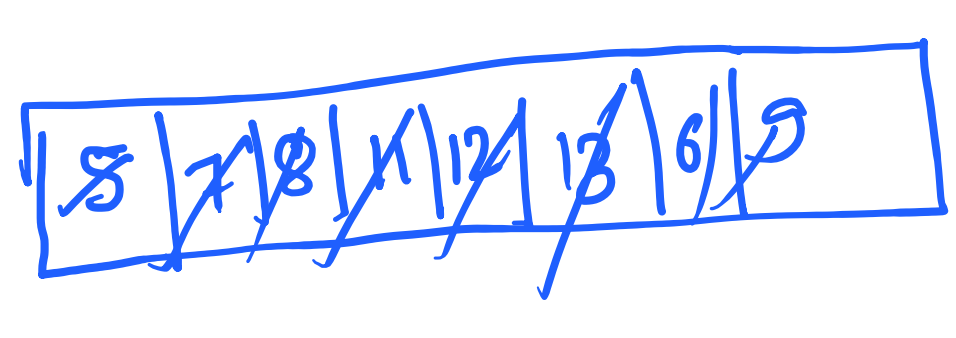


```
public void leveleOrder() {
    Queue<Node> q = new LinkedList<>();
    q.add(root);
    while(!q.isEmpty()) {
        Node n=q.poll();
        System.out.print(n.val+" ");
        if(n.left!=null) {
            q.add(n.left);
        }
        if(n.right!=null) {
            q.add(n.right);
        }
    }
    System.out.println();
}
```

0
5
1
2
7
8
11
12
13
6
9
7

5 7 8 11 12 13 6 9

[5, 7, 8, 11, 12, 13, 6, 9]


5
7
8
11
12
13
6
9